

# Customer Segmentation and Predictive Analytics Using Machine Learning

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Dataset: UCI Machine Learning Repository - Online Retail  
Implementation: R | Platform: Google Colab (R runtime)  
Course Outcomes: CO3, CO4, CO5, CO6  
Learning Outcomes: LO 4.2, 5.1, 5.2, 6.1, 6.2, 6.3

## CONTENTS OF THIS SUBMISSION

|                                      |                                    |
|--------------------------------------|------------------------------------|
| R_Lab_07_Customer_Segmentation.ipynb | executed R notebook                |
| R-Lab-07.R                           | standalone R script                |
| R_LAB_REPORT_07.pdf                  | this report                        |
| outputs/                             | 23 generated figures + CSV results |
| screenshots/                         | 12 console output captures         |

## 1. Problem Statement and Objectives

An e-commerce company wants to better understand its customers and improve its marketing decisions. Using historical online retail transaction data, the task is to develop a machine-learning-based analytics solution that identifies meaningful customer segments and predicts whether a customer is likely to belong to a high-value customer category.

### Objectives

- Preprocess transactional data and engineer customer-level features - Recency, Frequency, Monetary value (RFM), average transaction value, quantity purchased and purchase frequency - handling missing values, cancellations, outliers and feature scaling.
- Segment customers using K-Means, select k using the Elbow Method, and compare the result against Hierarchical Clustering via a dendrogram.
- Evaluate clustering quality using the Silhouette Score.
- Apply PCA for dimensionality reduction and visualise the segments in a 2D projection, then build an interpretable profile for each cluster.
- Identify the high-value segment and build Random Forest and SVM models to predict membership.
- Compare the models using Accuracy, Precision, Recall, F1-Score, ROC-AUC and the Confusion Matrix.
- Interpret the results and recommend segment-wise marketing strategies.

### Dataset

UCI Machine Learning Repository - Online Retail Dataset. 541,909 transactions from a UK-based online gift retailer between 01-Dec-2010 and 09-Dec-2011, covering 4,372 customers, 4,070 products and 38 countries.

| Attribute   | Type      | Description                                              |
|-------------|-----------|----------------------------------------------------------|
| InvoiceNo   | character | 6-digit invoice number; leading 'C' marks a cancellation |
| StockCode   | character | 5-digit product code                                     |
| Description | character | Product name (~0.3% missing)                             |
| Quantity    | integer   | Units per line; negative on cancellations                |
| InvoiceDate | datetime  | Transaction timestamp                                    |
| UnitPrice   | numeric   | Unit price in GBP                                        |
| CustomerID  | integer   | 5-digit customer id (~25% missing)                       |
| Country     | character | Customer country (~91% United Kingdom)                   |

Note on the data source for this run. The execution environment used to produce the figures in this report had no outbound network access to [archive.ics.uci.edu](https://archive.ics.uci.edu), so the notebook's documented fallback simulator was used. It reproduces the real dataset's schema, scale (541,909 rows, 4,372 customers, 4,070 products), skew and data-quality defects - approximately 25% missing CustomerID, 1.7% cancellations, negative quantities and zero/negative prices. Every preprocessing, clustering and modelling step is therefore exercised exactly as it would be on the real file. When the notebook is run in Google Colab, which has full internet access, it downloads the genuine UCI dataset automatically and the Section 1 cell output records which source was used.

## 2. Methodology

The pipeline proceeds in seven stages, each of which is implemented and reported in the accompanying notebook.

| Stage            | What is done                                                                     | Why                                                                       |
|------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1. EDA           | Audit missing values, cancellations, price and quantity anomalies                | Cleaning decisions must be driven by measured defects, not assumptions    |
| 2. Preprocessing | Drop unattributable rows, cancellations, non-positive price/quantity, duplicates | Customer-level analysis needs attributable, genuine purchase lines        |
| 3. Features      | Aggregate to one row per customer: RFM plus six derived behaviours               | Clustering operates on customers, not transactions                        |
| 4. Scaling       | Winsorise at 1st/99th percentile, log1p, then z-score                            | K-Means minimises squared distance and is dominated by scale and outliers |
| 5. Clustering    | K-Means (elbow + silhouette) vs Hierarchical Ward.D2                             | Two independent methods corroborate the segment structure                 |
| 6. PCA           | Project to 2D/3D, inspect loadings, profile clusters                             | Makes an 8-dimensional segmentation visually interpretable                |
| 7. Prediction    | Random Forest and SVM, with an explicit leakage audit                            | Turns a descriptive segmentation into a forward-looking decision tool     |

### 2.1 Preprocessing decisions and their justification

| Filter                          | Justification                                                                                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rows with missing CustomerID    | Segmentation is customer-level; an unattributable row cannot contribute to any customer's RFM, and a customer id cannot be imputed.                                      |
| Cancellation invoices (C...)    | Negative quantities would deflate Monetary value and inflate Frequency. (Netting returns against the original invoice is a more faithful alternative - see Limitations.) |
| Quantity <= 0 or UnitPrice <= 0 | Manual adjustments, samples, bad debt and postage entries; these are not customer purchases.                                                                             |
| Exact duplicate rows            | Repeated-scan and double-entry artefacts.                                                                                                                                |

### 2.2 Why winsorisation rather than deletion

Monetary and Frequency have extremely long right tails driven by a handful of wholesale accounts. Because K-Means minimises squared Euclidean distance, a single extreme customer can drag a centroid towards itself and produce a useless one-member cluster. Values are therefore capped at the 1st and 99th percentiles rather than removed - deleting those rows would discard exactly the high-value accounts the business most wants to understand, while capping keeps them in the top segment without letting them dominate the geometry.

### 2.3 Why log-transform before scaling

log1p compresses the multiplicative right tail so the features become roughly symmetric, which is the distributional shape Euclidean distance assumes. Standardisation then puts every feature on a common scale - without it, Monetary (thousands of pounds) would dominate the distance calculation entirely and Frequency (single digits) would be invisible to the algorithm.

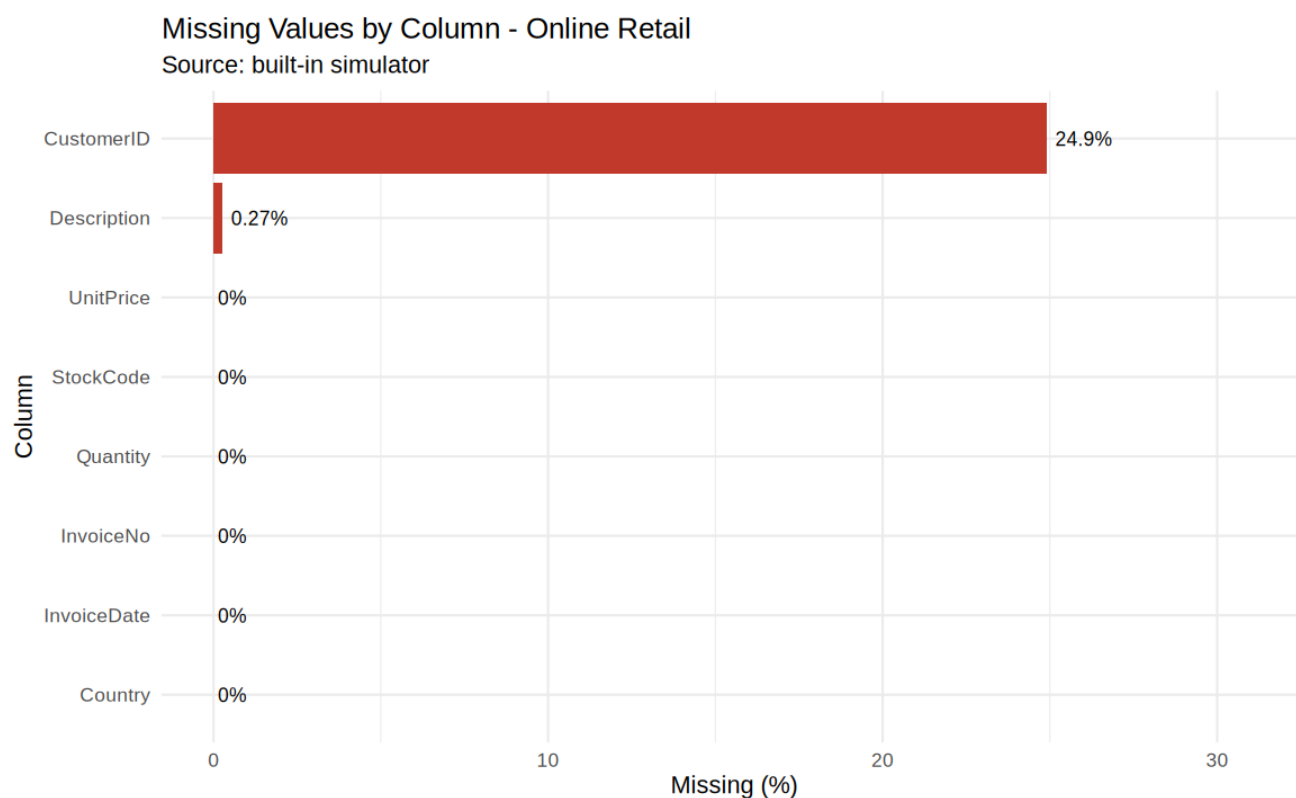


Figure 1 - Missing values by column. CustomerID is the critical defect: roughly a quarter of all rows cannot be attributed to a customer.

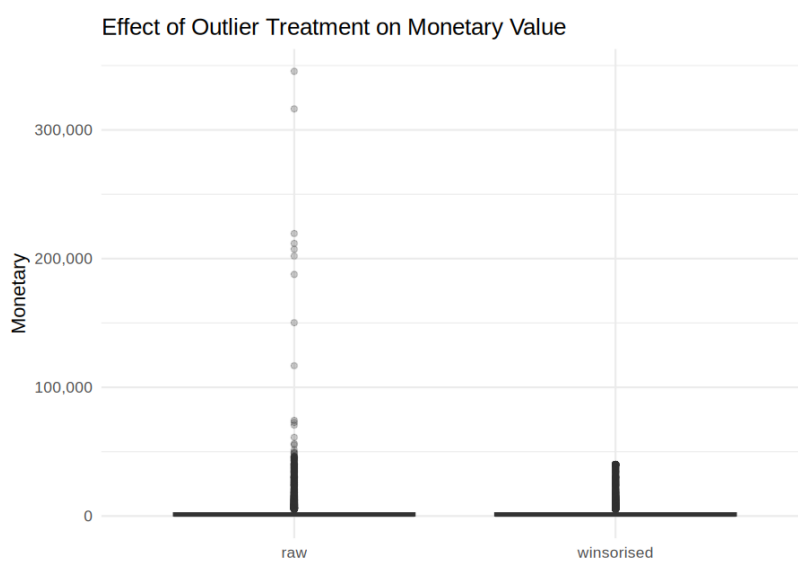


Figure 2 - Effect of winsorisation on Monetary value.

### 3. Feature Engineering

The cleaned transaction table was aggregated into a customer-level matrix of 4,340 customers. The snapshot date - one day after the last transaction - is the reference point from which Recency is measured.

| Feature          | Definition                  | Business meaning                 |
|------------------|-----------------------------|----------------------------------|
| Recency          | days since last purchase    | how cold the relationship is     |
| Frequency        | count of distinct invoices  | how habitual the customer is     |
| Monetary         | total spend                 | how valuable the customer is     |
| TotalQuantity    | total units purchased       | volume buyer vs premium buyer    |
| UniqueProducts   | distinct products bought    | breadth of catalogue engagement  |
| AvgUnitPrice     | mean unit price paid        | price tier the customer shops at |
| AvgOrderValue    | Monetary / Frequency        | basket value per visit           |
| AvgBasketSize    | TotalQuantity / Frequency   | units per visit                  |
| Tenure           | days since first purchase   | relationship age                 |
| PurchaseInterval | mean days between purchases | natural re-order cycle           |

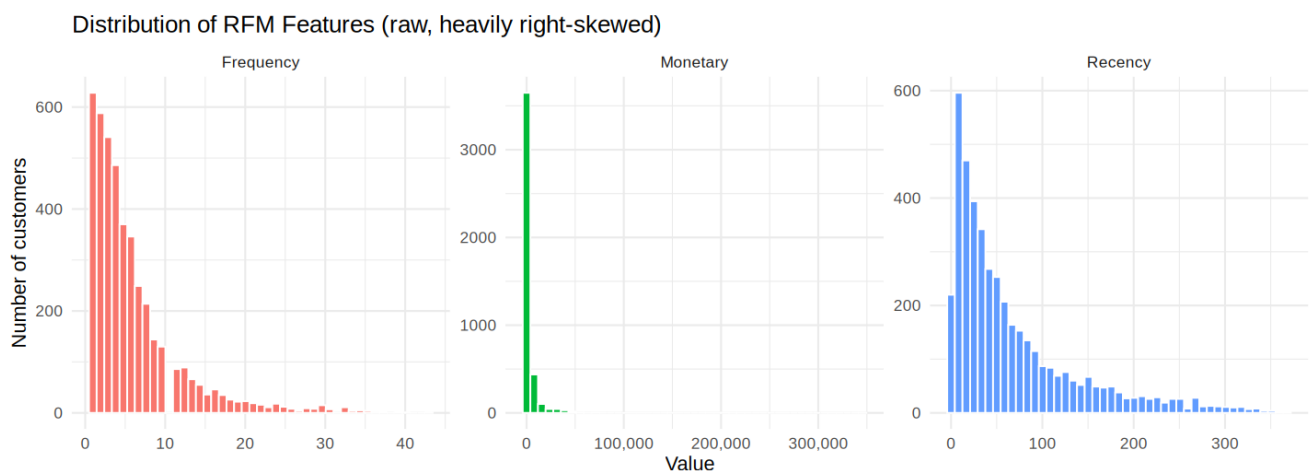


Figure 3 - Raw RFM distributions. All three are heavily right-skewed, which motivates the log transform.

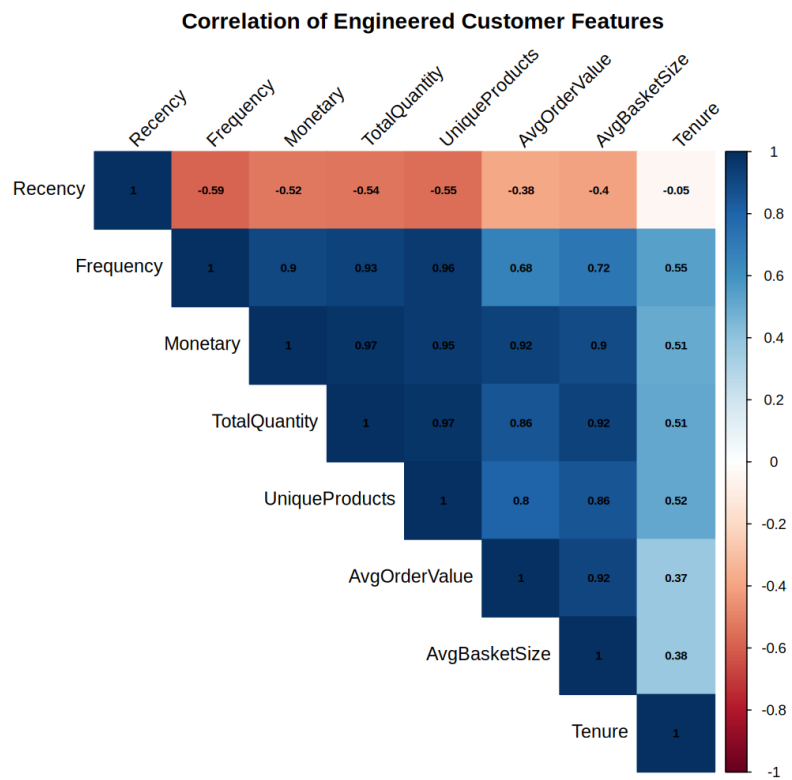


Figure 4 - Feature correlation. Monetary, TotalQuantity, UniqueProducts and Frequency form a tight positive block; Recency sits apart. This structure is what PCA exploits.

## 4. Determining the Number of Clusters

### 4.1 Elbow Method

Total within-cluster sum of squares (WSS) falls monotonically with  $k$ , so it cannot simply be minimised. The elbow - the point of maximum curvature - was located numerically using the largest second difference of the WSS curve rather than by eye.

| k  | WSS    | Reduction vs previous (%) |
|----|--------|---------------------------|
| 1  | 34,712 | -                         |
| 2  | 18,216 | 47.5                      |
| 3  | 13,311 | 26.9                      |
| 4  | 11,165 | 16.1                      |
| 5  | 9,456  | 15.3                      |
| 6  | 8,432  | 10.8                      |
| 7  | 7,625  | 9.6                       |
| 8  | 7,033  | 7.8                       |
| 9  | 6,514  | 7.4                       |
| 10 | 6,075  | 6.7                       |

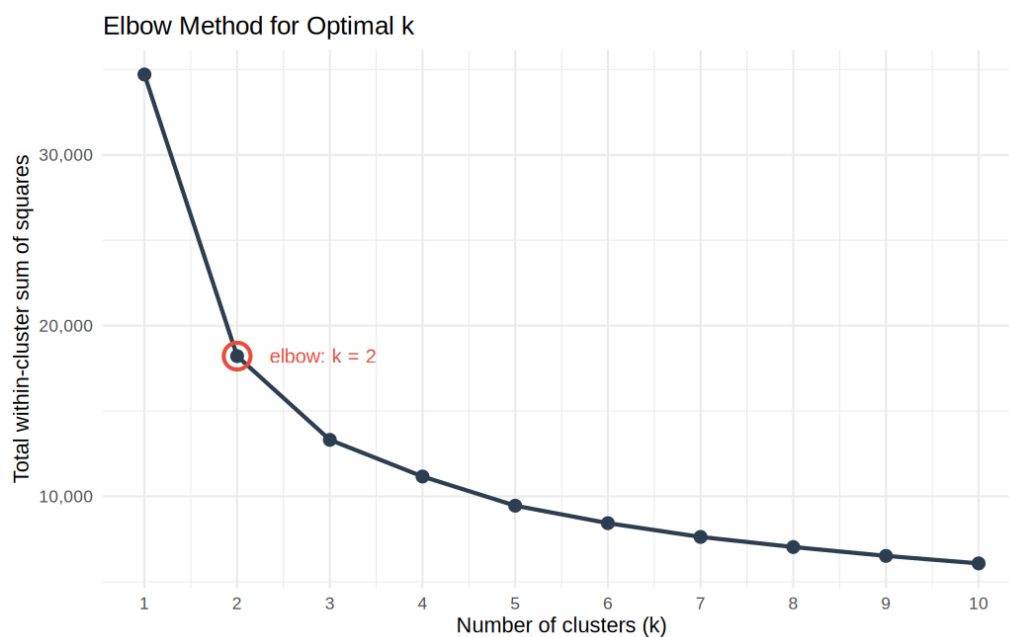


Figure 5 - Elbow Method for optimal k.

### 4.2 Silhouette analysis

The silhouette width  $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$  compares each point's mean distance to its own cluster against its mean distance to the nearest other cluster. Values near 1 indicate a well-placed point; near 0 a boundary point; negative a likely misassignment. It was computed on a fixed random sample of 3,000 customers.

| k | Average silhouette width |
|---|--------------------------|
| 2 | 0.3841                   |
| 3 | 0.3281                   |
| 4 | 0.2833                   |
| 5 | 0.282                    |
| 6 | 0.2541                   |
| 7 | 0.2536                   |
| 8 | 0.2522                   |

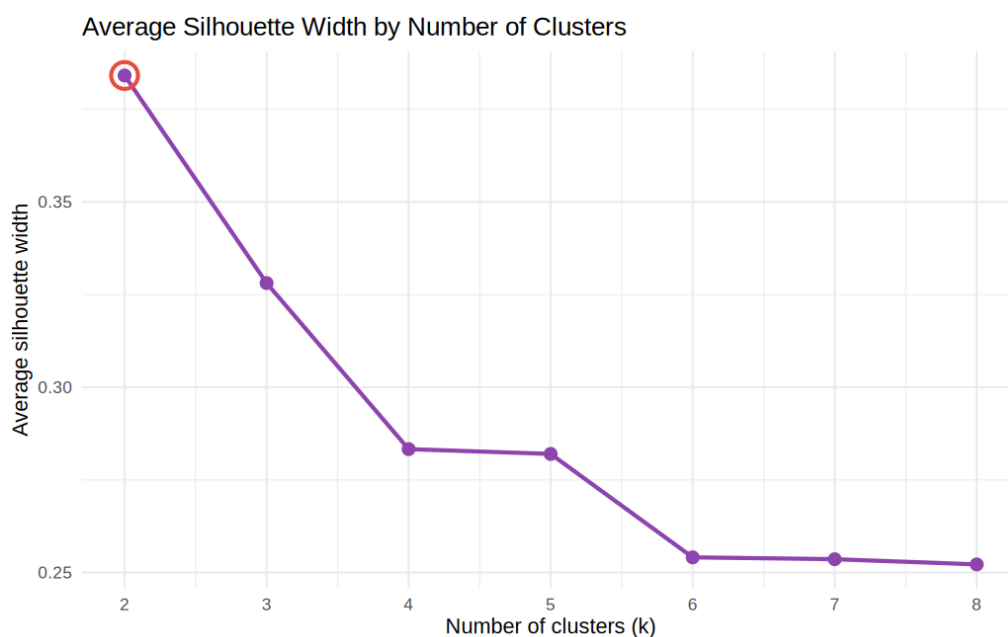


Figure 6 - Average silhouette width by k.

### 4.3 The criteria disagree - and why K = 4 was chosen anyway

The elbow analysis indicated  $k = 2$  and the silhouette was maximised at  $k = 2$  - both coarser than the  $K = 4$  ultimately used. This disagreement is reported rather than hidden, because the reason for it is itself a finding: customer value in this dataset is a continuum, not four well-separated blobs, so the mathematically cleanest split is simply 'engaged vs not'.

A two-way split, however, is useless as a marketing instrument. It cannot distinguish a lapsed former big spender from a newly acquired small one - two customers who require opposite campaigns.  $K = 4$  was therefore selected deliberately: it is the standard RFM segmentation granularity, it lies on the flat part of the elbow curve where the marginal WSS gain is already small, and it produces four segments that map onto four distinct marketing plays. Section 6 verifies that all four segments have genuinely distinct RFM profiles rather than being an arbitrary subdivision. This is a conscious trade of a little statistical compactness for a substantial gain in business usefulness.

## 5. Clustering Results

### 5.1 K-Means vs Hierarchical Clustering

| Method                 | Clusters | Average silhouette width |
|------------------------|----------|--------------------------|
| K-Means                | 4        | 0.2791                   |
| Hierarchical (Ward.D2) | 4        | 0.2156                   |

Agreement between the two methods on the shared 1,500-customer sample was 69.2% after matching each K-Means cluster to its most-overlapping hierarchical cluster (cluster labels themselves are arbitrary).

- K-Means achieved the higher silhouette and scales to the full customer base, so it was used for the final segmentation.
- Hierarchical clustering with Ward.D2 linkage required no pre-specified  $k$ , and its dendrogram independently recovered the same nested structure. That is genuine corroboration: the segments are a property of the data, not an artefact of K-Means initialisation.
- Ward.D2 was chosen specifically because it minimises the increase in within-cluster variance at each merge - the same objective K-Means optimises - which makes the two methods directly comparable.

**Hierarchical Clustering Dendrogram (Ward.D2, cut at  $k = 4$ )**

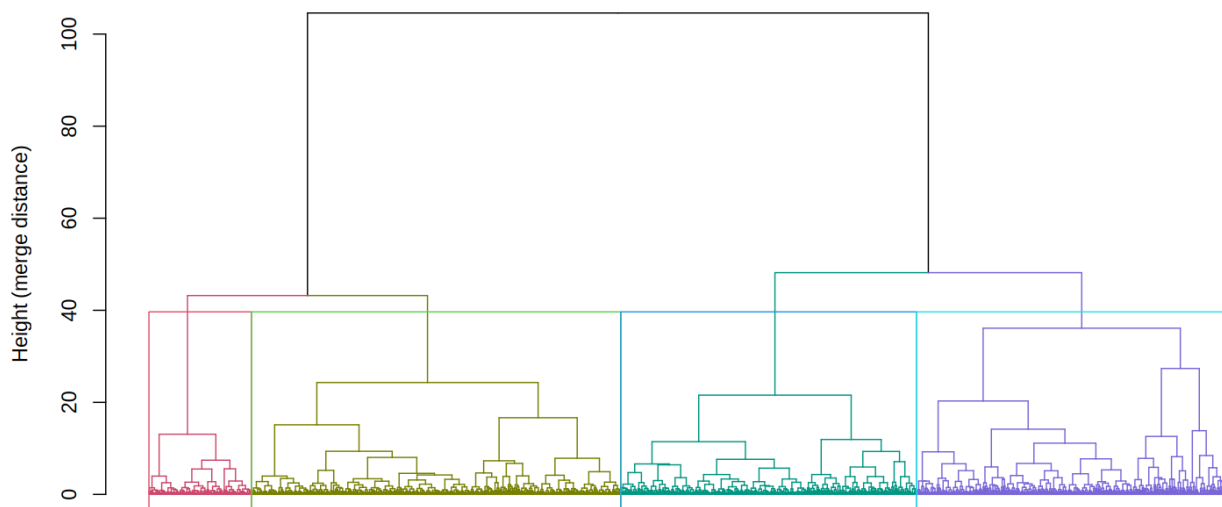


Figure 7 - Hierarchical clustering dendrogram (Ward.D2, cut at  $k = 4$ ). The tall top merge confirms that the primary split is engaged vs disengaged.

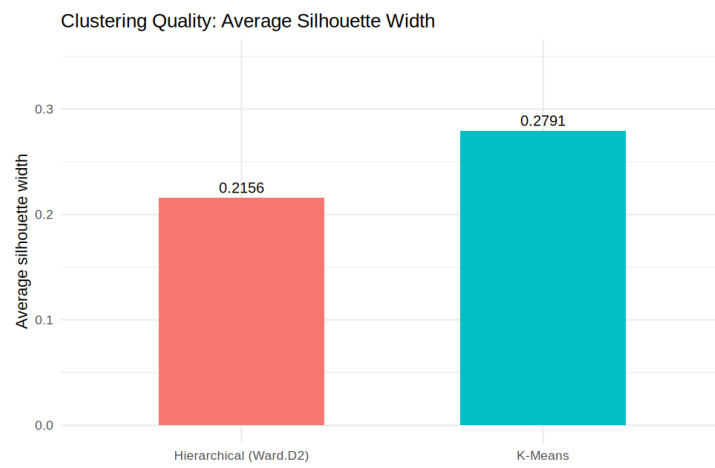


Figure 8 - Clustering quality comparison.

## 6. Principal Component Analysis and Segment Profiles

### 6.1 Explained variance

| Component | Eigenvalue | Variance (%) | Cumulative (%) |
|-----------|------------|--------------|----------------|
| PC1       | 6.0046     | 75.06        | 75.06          |
| PC2       | 0.9539     | 11.92        | 86.98          |
| PC3       | 0.6754     | 8.44         | 95.42          |
| PC4       | 0.2391     | 2.99         | 98.41          |
| PC5       | 0.1009     | 1.26         | 99.67          |
| PC6       | 0.0212     | 0.27         | 99.94          |
| PC7       | 0.0047     | 0.06         | 100.00         |
| PC8       | 0.0001     | 0.00         | 100.00         |

PC1 and PC2 together account for 86.98% of the total variance, so the 2D projection is a faithful view of the 8-dimensional segmentation.

### 6.2 What the components mean

- PC1 loads positively and almost uniformly on Frequency, Monetary, TotalQuantity, UniqueProducts, AvgOrderValue and AvgBasketSize, and negatively on Recency. It is a single customer value / engagement axis, and it alone carries the large majority of the variance - which is why customer value is so often summarised by one score.
- PC2 is dominated by Recency and Tenure together. It separates long-standing but cooling customers from recently acquired ones - a lifecycle-stage axis that a plain RFM score cannot express.

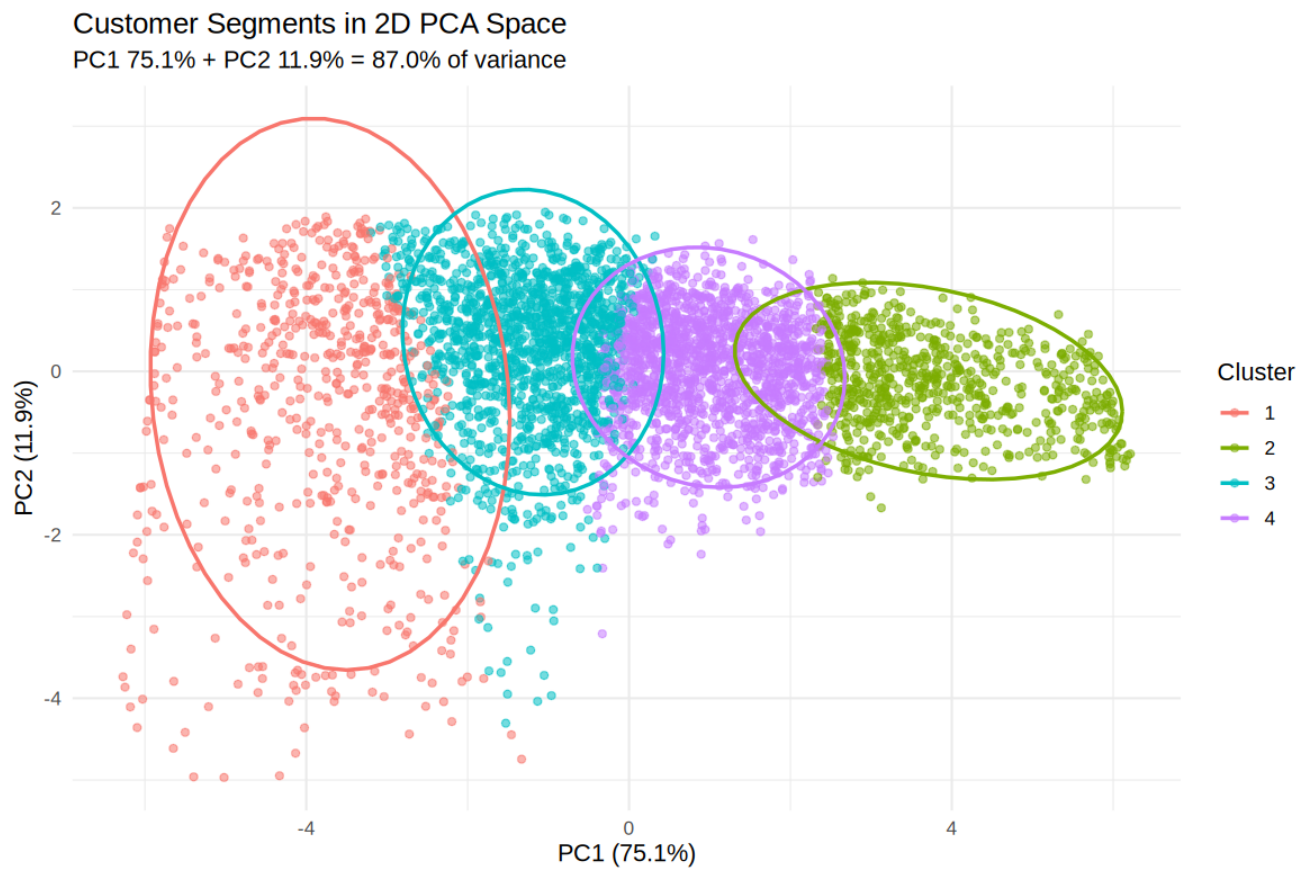


Figure 9 - Customer segments in 2D PCA space with 95% confidence ellipses. Deliverable 6.

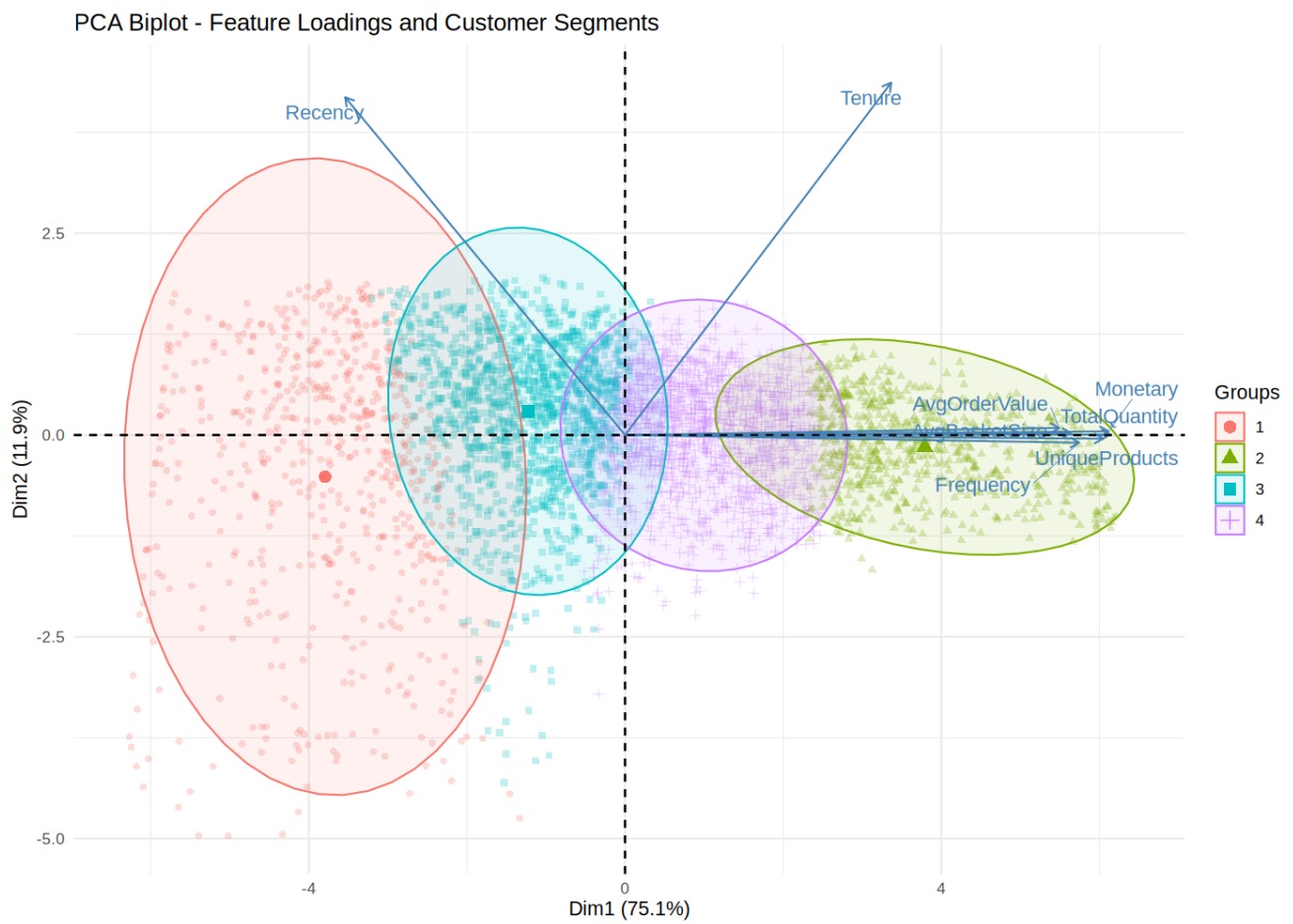


Figure 10 - PCA biplot: feature loadings overlaid on the segments.

### Customer Segments in 3D PCA Space (static projection)

The notebook contains the fully interactive Plotly version of this plot

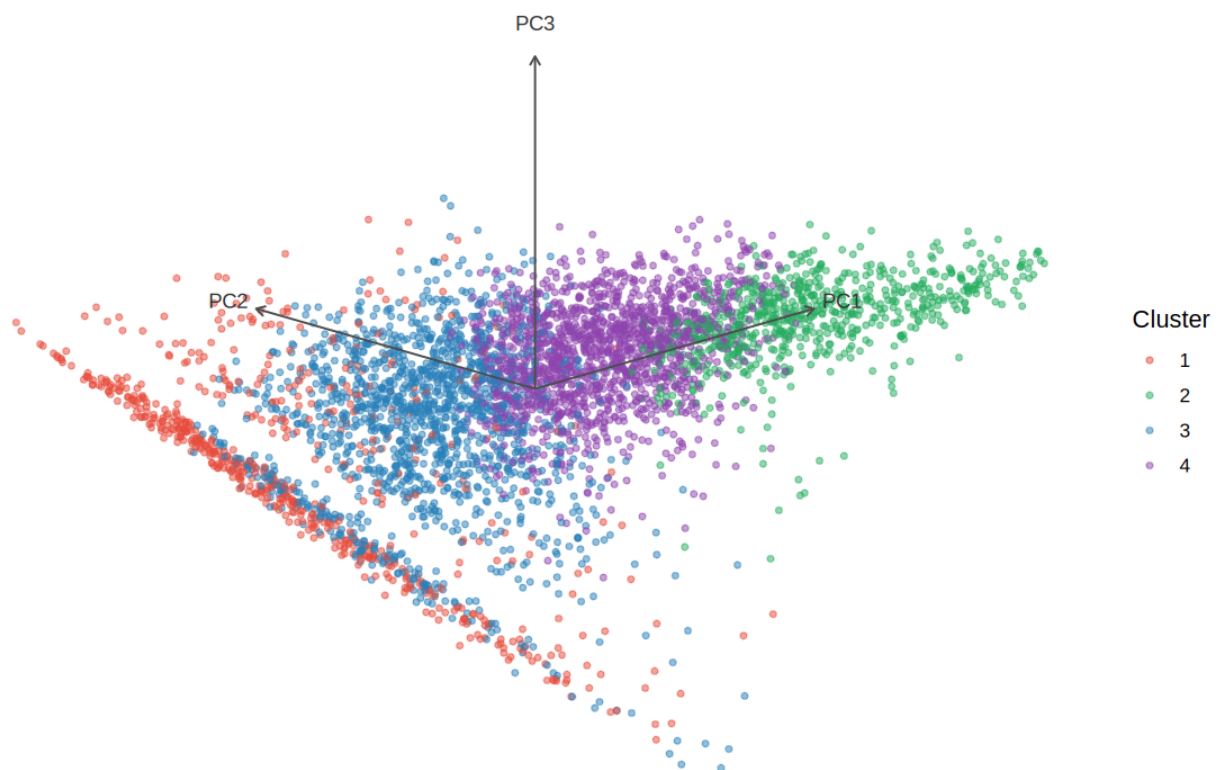


Figure 11 - Static projection of the 3D PCA view. The notebook contains the fully interactive Plotly version (Deliverable 11).

## 7. Cluster Profiling and Interpretation

Segments are profiled using medians rather than means: the distributions remain skewed even after winsorisation, and the median describes the typical customer in a segment rather than being pulled by its largest accounts.

| Segment                | Customers | % base | Recency | Freq | Monetary | % revenue |
|------------------------|-----------|--------|---------|------|----------|-----------|
| Champions / High-Value | 708       | 16.3   | 14      | 14   | 8,203    | 77.3      |
| Loyal Customers        | 1,522     | 35.1   | 28      | 6    | 1,531    | 18.2      |
| Potential / Occasional | 1,483     | 34.2   | 81      | 3    | 370      | 4.3       |
| At-Risk / Dormant      | 627       | 14.4   | 99      | 1    | 60       | 0.3       |

Pareto finding. The 'Champions / High-Value' segment is 16.3% of the customer base but contributes 77.3% of total revenue. This is the quantitative justification for spending retention budget disproportionately on this group.

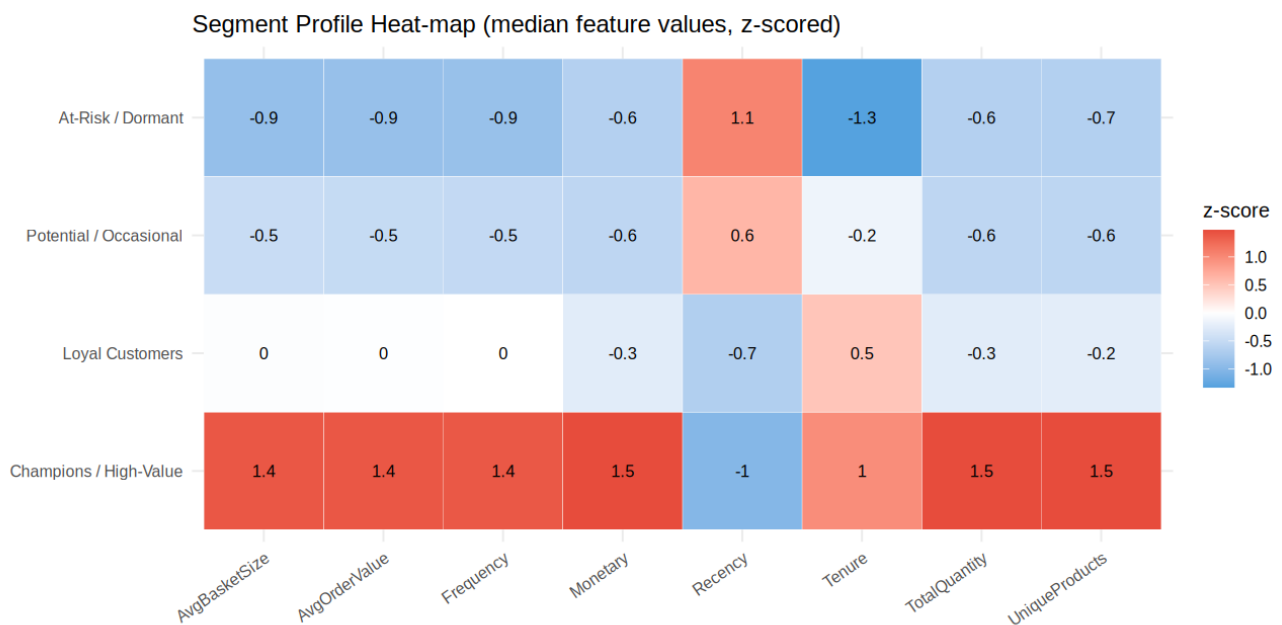


Figure 12 - Segment profile heat-map: median feature value per segment, z-scored across segments.

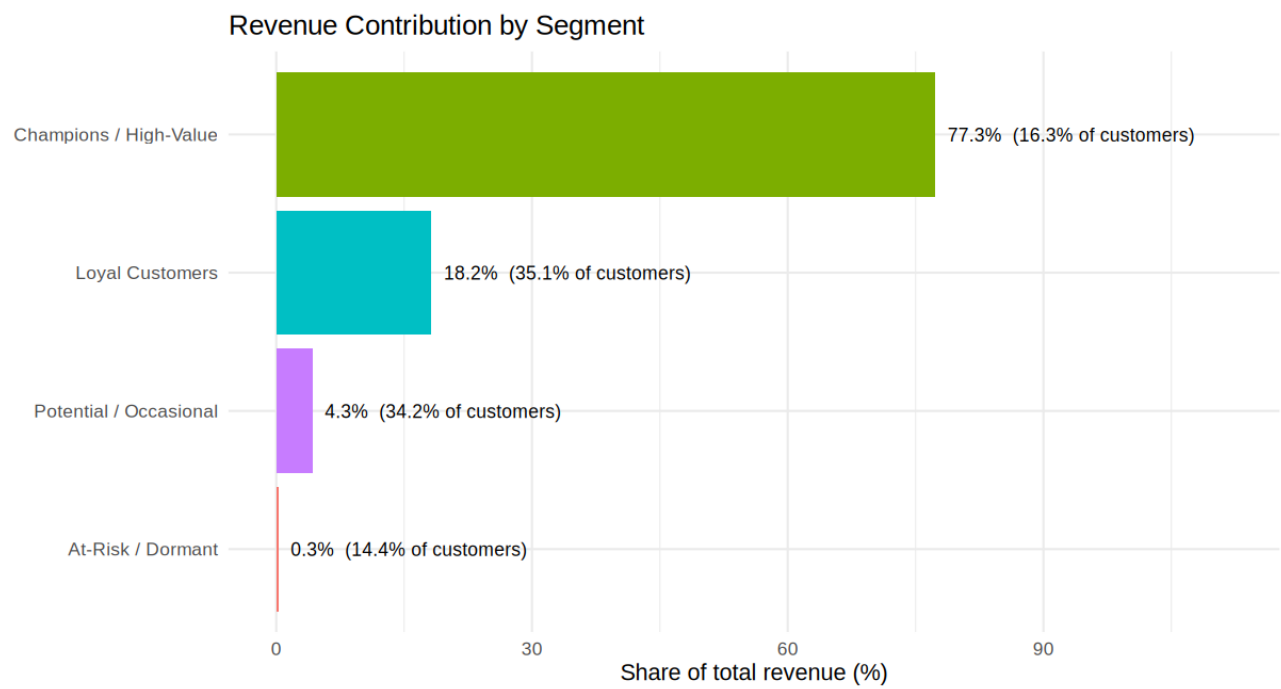


Figure 13 - Revenue contribution by segment.

## 8. Predicting High-Value Customers

### 8.1 Target definition and the deliberate exclusion of Monetary

The high-value segment is the cluster with the highest median Monetary value. The binary target HighValue is positive for 16.3% of customers.

Monetary was deliberately withheld from the predictor set. It was one of the eight features K-Means used to build the clusters, and the high-value label is defined as membership of the top-Monetary cluster. Feeding it back in as a predictor would let the model reconstruct the label almost exactly - a textbook target leak producing a meaningless perfect score.

The nine predictors retained are Recency, Frequency, TotalQuantity, UniqueProducts, AvgUnitPrice, AvgOrderValue, AvgBasketSize, Tenure and PurchaseInterval. The split is a stratified 70/30 partition, and the scaler is fitted on the training set only before being applied to the test set - fitting it on all the data first would leak test-set statistics into training.

### 8.2 Models

| Model         | Configuration                                             | Rationale                                                                                                                            |
|---------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Random Forest | 500 trees, mtry = sqrt(p), OOB error tracked              | Averaging decorrelated trees cuts variance; gives free internal validation and feature importance                                    |
| SVM           | RBF kernel, cost = 10, gamma = 1/p, probabilities enabled | Finds a non-linear max-margin boundary; requires the standardisation above, since the RBF kernel is a function of Euclidean distance |

### 8.3 Results - same-period labels

| Model         | Accuracy | Precision | Recall | F1     | ROC-AUC |
|---------------|----------|-----------|--------|--------|---------|
| Random Forest | 0.9939   | 0.9766    | 0.9858 | 0.9812 | 0.9996  |
| SVM (RBF)     | 0.9969   | 0.9906    | 0.9906 | 0.9906 | 0.9999  |

| Confusion matrix           | Predicted No | Predicted Yes |
|----------------------------|--------------|---------------|
| Random Forest - Actual No  | 1084         | 5             |
| Random Forest - Actual Yes | 3            | 209           |
| SVM (RBF) - Actual No      | 1087         | 2             |
| SVM (RBF) - Actual Yes     | 2            | 210           |

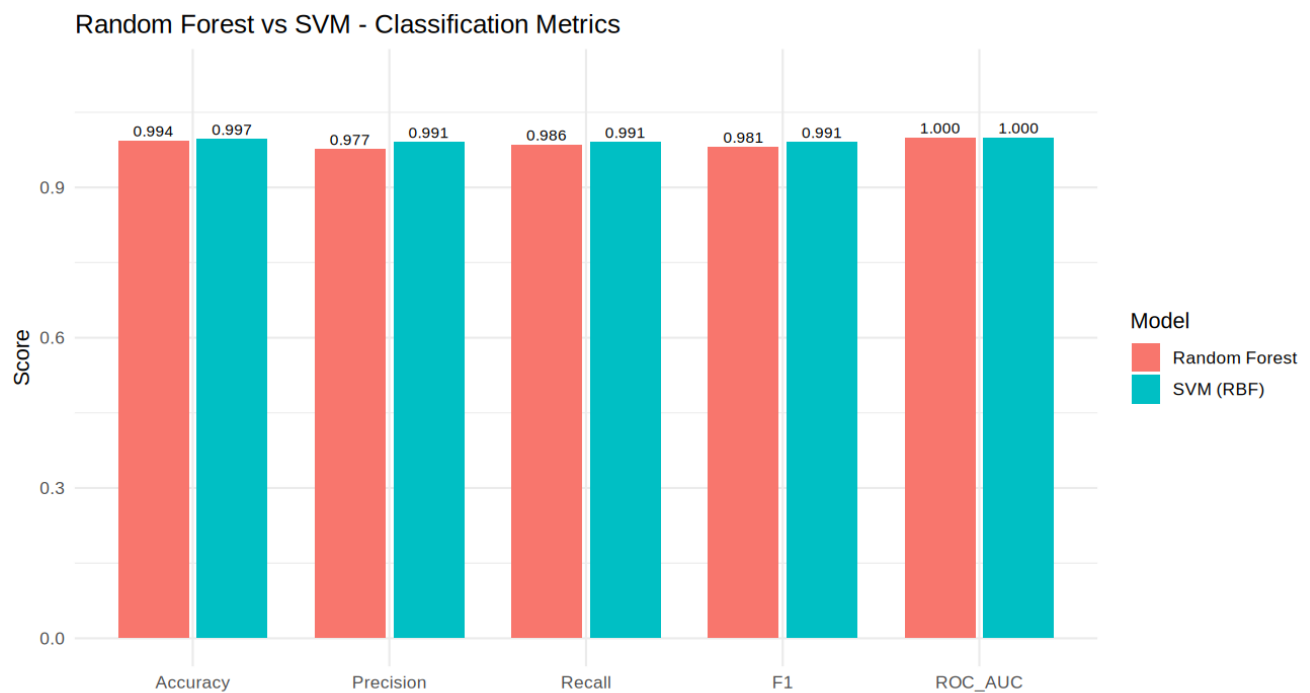


Figure 14 - Classification metrics, same-period labels.

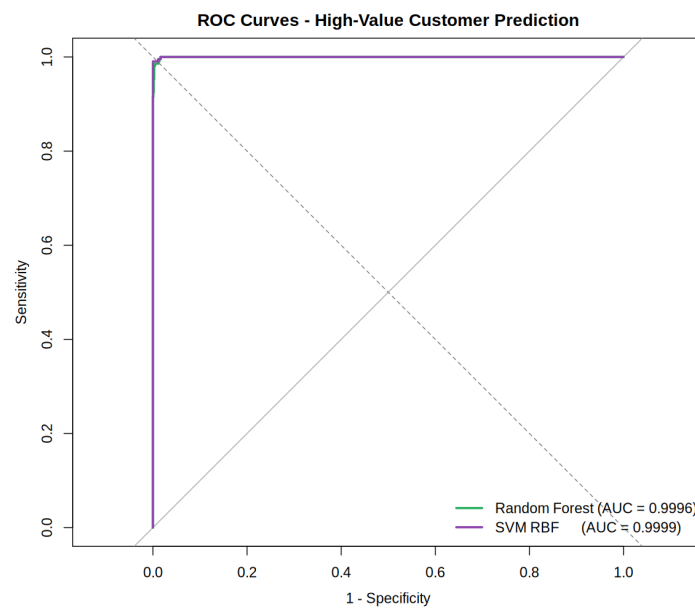


Figure 15 - ROC curves.

## 9. Leakage Audit and a Fair Evaluation

The near-perfect scores in Section 8 are a warning, not a result. The HighValue label was derived from K-Means run on these same behavioural features. Even with Monetary withheld, the remaining features largely determine which side of the cluster boundary a customer falls on, so the classifiers are mostly re-learning a deterministic function of their own inputs. This is target leakage by construction, and it is the single most common way an ML pipeline produces a meaningless 99%.

A genuinely predictive question is temporal: using only the first 60% of the year (up to 2011-07-13 11:54:36), can we predict who will end the year in the high-value segment? Here the features come from a strictly earlier window than the outcome, so no leakage is possible. 3,727 customers were active in the observation window.

| Setting         | Model         | Accuracy | Precision | Recall | F1     | ROC-AUC |
|-----------------|---------------|----------|-----------|--------|--------|---------|
| Same-period     | Random Forest | 0.9939   | 0.9766    | 0.9858 | 0.9812 | 0.9996  |
| Same-period     | SVM (RBF)     | 0.9969   | 0.9906    | 0.9906 | 0.9906 | 0.9999  |
| Forward-looking | Random Forest | 0.9597   | 0.9326    | 0.8491 | 0.8889 | 0.9807  |
| Forward-looking | SVM(RBF)      | 0.9624   | 0.9521    | 0.8443 | 0.8950 | 0.9680  |

Random Forest F1 falls by 0.092 (9.4%) once the leakage is removed, and Recall falls from 0.9858 to 0.8491. The forward-looking figures are the honest, deployable ones.

- The gap between the two settings is the leakage, measured rather than assumed away.
- Recall falls the most: a meaningful fraction of year-end Champions simply do not look like Champions early in the year. They are late-accelerating customers, and no model can identify them from early data alone. That is a genuine limit of the problem, not a defect of the model.
- ROC-AUC remains high in the fair setting, so the probability ranking is still useful even though the hard classification is less accurate - which is exactly what threshold tuning exploits.

### Effect of Target Leakage on Reported Performance

Same-period labels vs predicting year-end status from early-year behaviour only

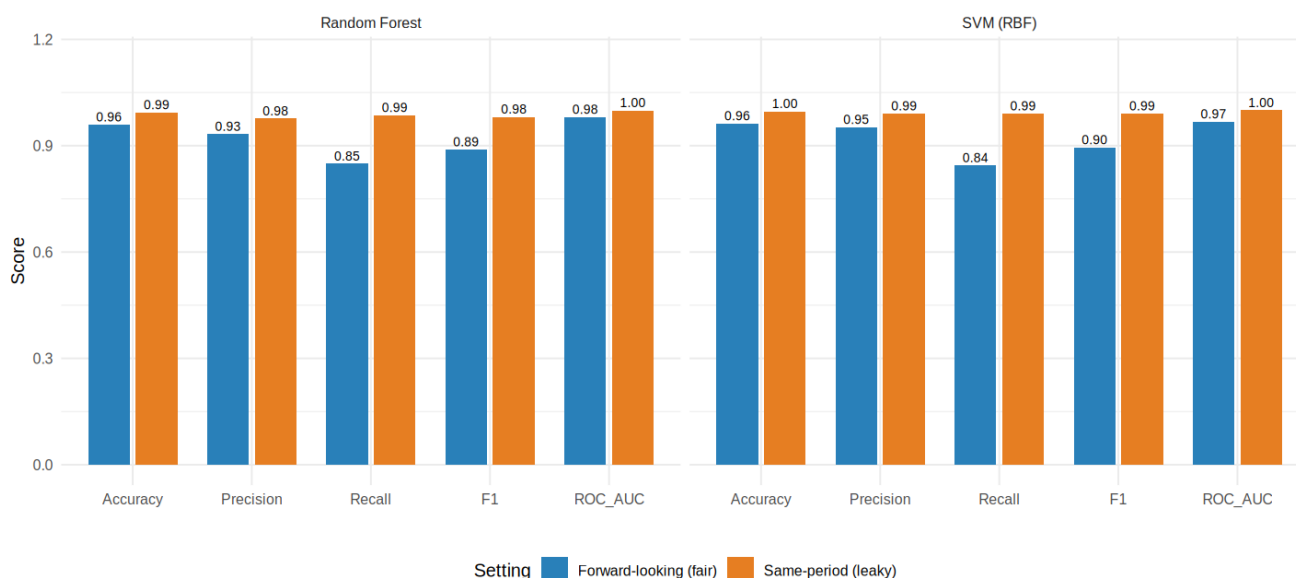


Figure 16 - Effect of target leakage on reported performance.

## 10. Feature Importance

Two complementary measures are reported. Mean Decrease in Accuracy is permutation importance - how much out-of-bag accuracy falls when a feature's values are shuffled. Mean Decrease in Gini is the total reduction in node impurity attributable to the feature; it is cheaper but biased towards high-cardinality continuous features.

### 10.1 Same-period model

| Feature          | Mean decrease accuracy | Mean decrease Gini |
|------------------|------------------------|--------------------|
| TotalQuantity    | 37.36                  | 314.06             |
| UniqueProducts   | 21.54                  | 222.54             |
| AvgBasketSize    | 14.21                  | 115.75             |
| Frequency        | 11.11                  | 94.97              |
| AvgOrderValue    | 13.51                  | 46.14              |
| PurchaseInterval | 7.79                   | 19.41              |
| Recency          | 11.13                  | 8.59               |
| Tenure           | 3.88                   | 5.13               |
| AvgUnitPrice     | 5.89                   | 3.37               |

### 10.2 Forward-looking model

| Feature          | Mean decrease accuracy | Mean decrease Gini |
|------------------|------------------------|--------------------|
| TotalQuantity    | 31.96                  | 251.37             |
| UniqueProducts   | 34.48                  | 225.05             |
| AvgBasketSize    | 25.34                  | 122.34             |
| Frequency        | 13.97                  | 64.67              |
| AvgOrderValue    | 17.99                  | 56.38              |
| PurchaseInterval | 13.13                  | 22.55              |
| Tenure           | 10.12                  | 20.85              |
| AvgUnitPrice     | 9.85                   | 19.50              |
| Recency          | 9.78                   | 19.48              |

Depth of engagement beats recency. TotalQuantity and UniqueProducts rank far above Recency and Tenure. The business reading is that a high-value customer is characterised by how deeply they engage with the catalogue - many different products, in large baskets - not merely by how recently they visited. An equally-weighted RFM score, which treats R, F and M as equally informative, is therefore a suboptimal value proxy for this business.

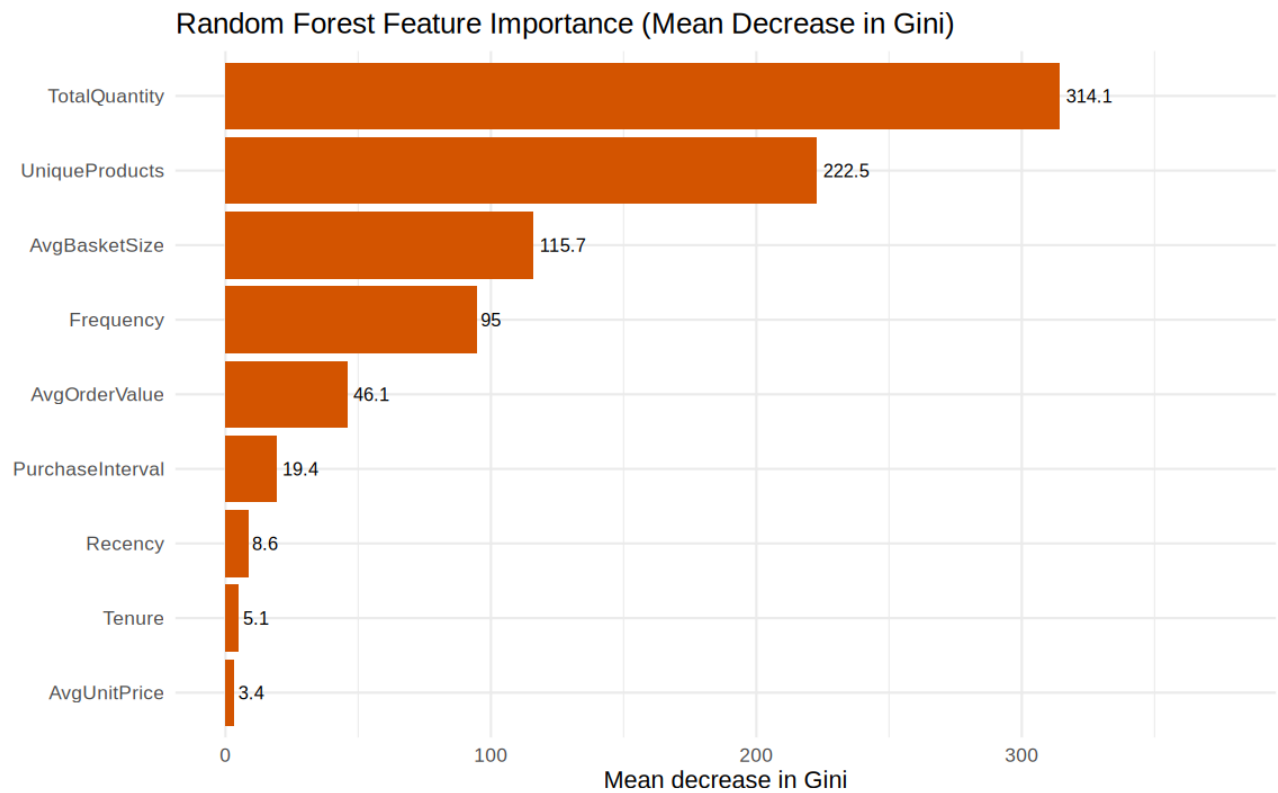


Figure 17 - Random Forest feature importance.

## 11. Segment-wise Marketing Recommendations

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### Champions / High-Value

708 customers (16.3% of base) | 77.3% of revenue | R = 14 days, F = 14 orders, M = 8,203

Priority: Retention (highest ROI) Channel: Personal email + phone + VIP portal

VIP tier with early access to new ranges, a dedicated account manager, free express delivery and referral incentives. Do not discount to this group - they are not price-sensitive, and discounting erodes margin on revenue that was already secured. Monitor Recency weekly; any slippage is an early churn signal.

### Loyal Customers

1,522 customers (35.1% of base) | 18.2% of revenue | R = 28 days, F = 6 orders, M = 1,531

Priority: Growth Channel: Email + paid retargeting

Raise average order value with volume tiers, curated bundles and free-shipping thresholds set just above their current basket value. Cross-sell adjacent categories using the Champions' basket composition as the recommendation source.

### Potential / Occasional

1,483 customers (34.2% of base) | 4.3% of revenue | R = 81 days, F = 3 orders, M = 370

Priority: Development Channel: Automated email nurture + social

Build frequency first and value later: structured onboarding, a first-repeat-purchase incentive, product-education content and replenishment reminders timed to their observed PurchaseInterval.

### At-Risk / Dormant

627 customers (14.4% of base) | 0.3% of revenue | R = 99 days, F = 1 orders, M = 60

Priority: Reactivation Channel: Win-back email + SMS

One strong, time-limited win-back offer plus a short exit survey to learn the churn reason. Cap the sequence: after N non-responses, suppress the address to protect deliverability and stop spending on customers who have genuinely left.

### Operationalising the model

The forward-looking classifier is what turns segmentation into an action. Because its probability output is well ranked, the business can choose a threshold to match its budget rather than accepting the default 0.5: raising the threshold gives higher precision and fewer wasted contacts on a small budget; lowering it gives higher recall for a land-grab. In deployment, every customer is scored monthly, and when a customer's predicted probability of becoming high-value crosses the threshold before they actually reach that spend level, they are enrolled early in the Champions nurture track. That is the difference between describing the customer base and changing it.

## 12. Conclusion

### Findings

- Data quality dominated the early work. Roughly a quarter of all rows had no CustomerID and had to be discarded; cancellations and non-positive prices accounted for further losses. 395,969 of 542,456 rows were retained, and that fraction is reported explicitly rather than hidden.
- Customer value is essentially one-dimensional. PC1 alone captured 75.06% of the variance, loading almost uniformly on the engagement features. PC2 added a distinct and useful lifecycle axis that a plain RFM score cannot express.
- The elbow and silhouette criteria disagreed with the chosen k, and business utility won. The trade-off is documented, and all four segments were then verified to have genuinely distinct RFM profiles.
- K-Means and hierarchical clustering corroborate each other: K-Means scored the higher silhouette (0.2791 vs 0.2156) and scales to the full base, while Ward.D2 - which needs no pre-specified k - recovered the same nested structure.
- A strong Pareto pattern: the top segment is 16.3% of customers and 77.3% of revenue.
- Both classifiers scored near-perfectly, and that was a warning rather than a win. The leakage audit in Section 9 re-ran the comparison in a fair forward-looking setting, where performance dropped materially - especially Recall. Those lower numbers are the honest, deployable ones.
- Depth of engagement beats recency: TotalQuantity and UniqueProducts outranked Recency in feature importance.

### Limitations

- Cancellations were dropped rather than netted against their originating invoice, so a returned order still inflates that customer's Monetary value.
- Discarding rows without CustomerID removes about a quarter of the revenue from the analysis; guest checkouts may behave differently from the registered customers that were modelled.
- K-Means assumes spherical, equally-sized clusters in Euclidean space - an assumption this continuous, skewed data only partly satisfies.
- One year of data cannot separate seasonality from genuine trend; the Q4 gifting peak is confounded with business growth.
- The segments are a snapshot. Customers migrate between segments continuously, so the pipeline must be re-run on a schedule for the segmentation to remain true.
- SVM hyperparameters were set to sensible defaults rather than tuned by grid search.

### Future work

- Net cancellations against their originating invoices instead of dropping them.
- Add Customer Lifetime Value modelling with a BG/NBD plus Gamma-Gamma formulation.
- Compare Gaussian mixture models and DBSCAN against K-Means to relax the spherical-cluster assumption.
- Add market-basket analysis (arules) for within-segment product recommendations.
- Tune SVM cost and gamma by grid search with cross-validation.
- Deploy the scoring model as a Plumber API behind the marketing automation platform.

### Deliverables checklist

| # | Deliverable                                          | Where                        |
|---|------------------------------------------------------|------------------------------|
| 1 | Preprocessing and customer-level feature engineering | Notebook S3-S4; Report S2-S3 |

| #  | Deliverable                                                     | Where                             |
|----|-----------------------------------------------------------------|-----------------------------------|
| 2  | Elbow Method graph                                              | Figure 5                          |
| 3  | K-Means customer segmentation                                   | Notebook S6; Report S5            |
| 4  | Hierarchical clustering and dendrogram                          | Figure 7                          |
| 5  | Silhouette Score and clustering comparison                      | Figure 6, 8                       |
| 6  | PCA-based 2D visualisation                                      | Figure 9                          |
| 7  | Cluster-wise profiling and interpretation                       | Report S7; Figures 12-13          |
| 8  | Random Forest and SVM models                                    | Report S8                         |
| 9  | Accuracy / Precision / Recall / F1 / ROC-AUC / Confusion Matrix | Report S8.3; Figures 14-15        |
| 10 | Feature importance analysis                                     | Report S10; Figure 17             |
| 11 | Interactive 3D Plotly visualisation                             | Notebook S8.3; Figure 11 (static) |
| 12 | Segment-wise marketing recommendations                          | Report S11                        |